Vysakh Ramakrishnan

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PROFILE SUMMARY

Full-stack AI engineer with experience across backend systems, ML deployment, and data pipelines. Strong background in TypeScript, Node.js, Next.js, and React, building scalable microservices, RESTful APIs, and productionizing AI systems. Skilled in PostgreSQL, AWS, Docker, and automation/testing.

PROFESSIONAL EXPERIENCE

Sony Europe Limited — **June 2025 – November 2025**
AI Engineer Intern — *Stuttgart, Germany*

- Developed **AI agent orchestration infrastructure** for real-time voice systems, coordinating STT, LLM reasoning, and TTS synthesis with streaming responses and latency-aware pipeline control.
- Designed and implemented a **temporally-aware semantic retrieval service** using TypeScript and Node.js with scalable backend components, RESTful API boundaries, and automated evaluation workflows.
- Developed a **differentiable ML framework** integrating physical simulation with neural reconstruction, contributing to production-grade pipelines, monitoring, and performance optimization across SLURM distributed systems.

Kontext.dev — **September 2025 – October 2025**
Software Engineer (AI / Full Stack) — *Freelance, Germany (Remote)*

- Built and deployed a **semantic retrieval system** over **Turbopuffer** using TypeScript and Node.js, implementing vector indexing, metadata-normalized schemas, and query-time filtering, reducing end-to-end inference latency by 50%.
- Designed and integrated **Vault-backed** information stores into the SDK, enabling secure document ingestion, structured metadata management, and consistent retrieval flows across downstream AI services.
- Developed a **real-time avatar streaming** web application using Next.js, TypeScript, and React with LiveKit WebRTC integration, server-side JWT token generation via RESTful APIs, and WebSocket-based media transport for interactive virtual presence demos.
- Implemented end-to-end retrieval pipelines including embedding generation, index lifecycle management, and API-layer orchestration, with CI-driven testing and containerized deployments using Docker and AWS.

Massachusetts General Hospital, Harvard Medical School — **August 2023 – March 2024**
Research Scholar, Advised By Prof.Sandeep Manjanna, Dr.Lana Schumacher — *Boston, Massachusetts, USA*

- Developed **real-time segmentation and object tracking pipelines** for robotic surgery videos, achieving 92% IoU and 81–89% IoU for multi-class anatomy segmentation using our memory-based **Transformer** model.
- Built a **Visual Question Localized-Answering (VQLA)** module to detect and localize key anatomic features in surgical scenes, integrating the multi-modal system into a query-driven interface.

Capgemini Technology Services India Limited — **September 2019 – May 2022**
Senior Full-Stack Engineer — *Bangalore, India*

- Implemented end-to-end features across SAP UI, ABAP business logic, and SQL-backed tablespaces, delivering production workflows used daily by finance teams.
- Traced data end-to-end (**screen → ABAP logic → database**), fixed ledger mismatches via SAP Notes and targeted data corrections, and improved reliability for high-volume transactions.

PROJECTS

Malayalam Voice Agent for Customer care support | *FastAPI, WebSockets, VAD, STT, LLM, TTS, Vitest*

- Built a full-duplex Malayalam voice assistant (VAD→STT→LLM→TTS) with barge-in cancellation under 400 ms, latency-masking fillers, and tone-aware responses over a validated WebSocket API.
- Built event-driven cancellation system tracking TTS playback state with per-chunk timestamps, triggering instant cancellation (< 400 ms target) when user speech detected, measuring and reporting stop latency metrics for every interruption event.
- Developed supporting tooling including a REST TTS service with browser demo and a load harness simulating 20 concurrent calls to measure TTFB and barge-in stop times against < 900 ms / < 400 ms targets.

Computer Use Backend | *FastAPI, SSE, SQLite, Docker*

- Wrapped Anthropic's computer-use agent loop in a session-centric FastAPI service with per-session locking, durable chat history, and persisted tool outputs for reliable multi-turn automation.
- Streamed assistant deltas and tool results over SSE while packaging a lightweight HTML/JS demo with embedded noVNC using Docker Compose for rapid local validation.

Contract Clause Analyzer | *FastAPI, React 18, TypeScript, Node.js, Postgres, Chroma, Docker, Vitest*

- Built a guardrailed contract clause analysis system that extracts clauses, retrieves playbook context via Chroma RAG, scores deviations, and streams results over SSE through a FastAPI backend and React UI.
- Architected immutable playbook versioning system with ChromaDB vector store, 800-word sliding window chunking, grounding enforcement requiring chunk citations for every finding, and hot reindexing without downtime.
- Hardened the pipeline with prompt-injection defenses, strict Pydantic schema validation, grounding checks, per-IP rate limiting, and token usage/cost tracking backed by Postgres/SQLite.

EDUCATION

Ecole Polytechnique, France

Masters (M2) in Artificial Intelligence and Advanced Visual Computing; CGPA: 3.79

- *Relevant Courses:* NLP, Deep Reinforcement Learning, Computer Graphics, Computer Vision

Plaksha University

Post Graduate Diploma in AI, ML, and Leadership; CGPA: 8.98

- *Relevant Courses:* Machine Learning, NLP, Computer Vision

IT SKILLS

Backend/Full Stack: TypeScript, Node.js, Next.js, React, Python, FastAPI, RESTful APIs, MongoDB, PostgreSQL, Redis, GraphQL

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Numpy, OpenCV.

Tools: Docker, AWS, Git, GitHub Actions, Vitest, SQLAlchemy, LangChain, LangGraph, ChromaDB, Prometheus